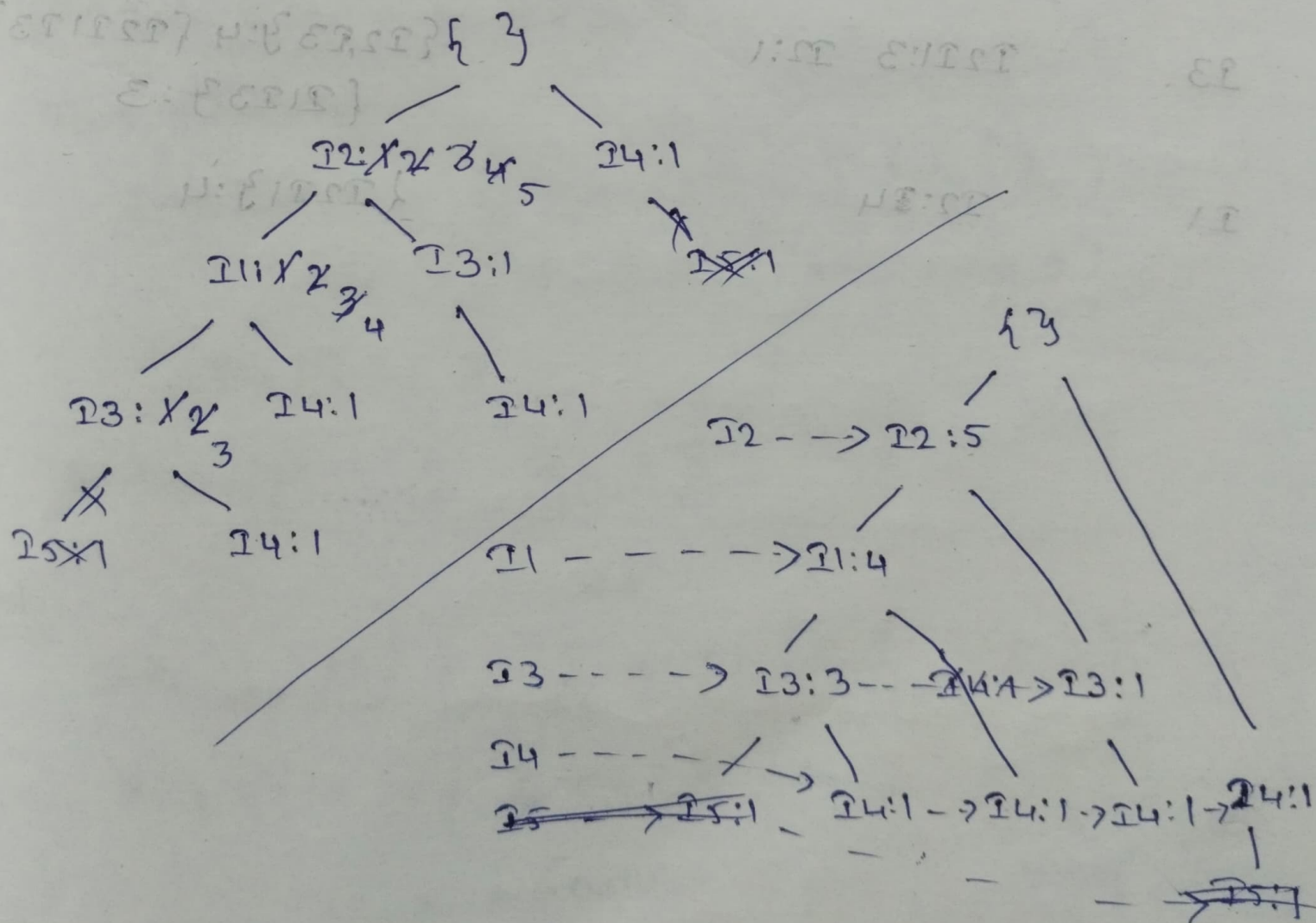


① Given : min-sup = 3.
Support threshold = 50%.

Support threshold = 50%.

Transactions	Items	Ordered items.
T1	I1, I2, I3	I2, I1, I3.
T2	I2, I3, I4	I2, I3, I4
T3	I4, I5	I4, I5
T4	I1, I2, I4	I2, I1, I4
T5	I1, I2, I3, I5	I2, I1, I3, I5
T6	I1, I2, I3, I4	I2, I1, I3, I4

$I_1 - 4, I_2 - 5, I_3 - 4, I_4 - 4, I_5 - 2 \times$



Item

Conditional
Pattern Base

Frequent Patterns Generated

I4

$I2I1I3:1$ $I2I3:1$ $I1I1:1$

$\{I2I4\}:3$, $\{I1I2I4\}:2$,
 $\{I2I3I4\}:2$, $\{I1I4\}:2$
 $\{I3I4\}:2$

Since min-support = 2.

$\{I2I4\}:3$

I3

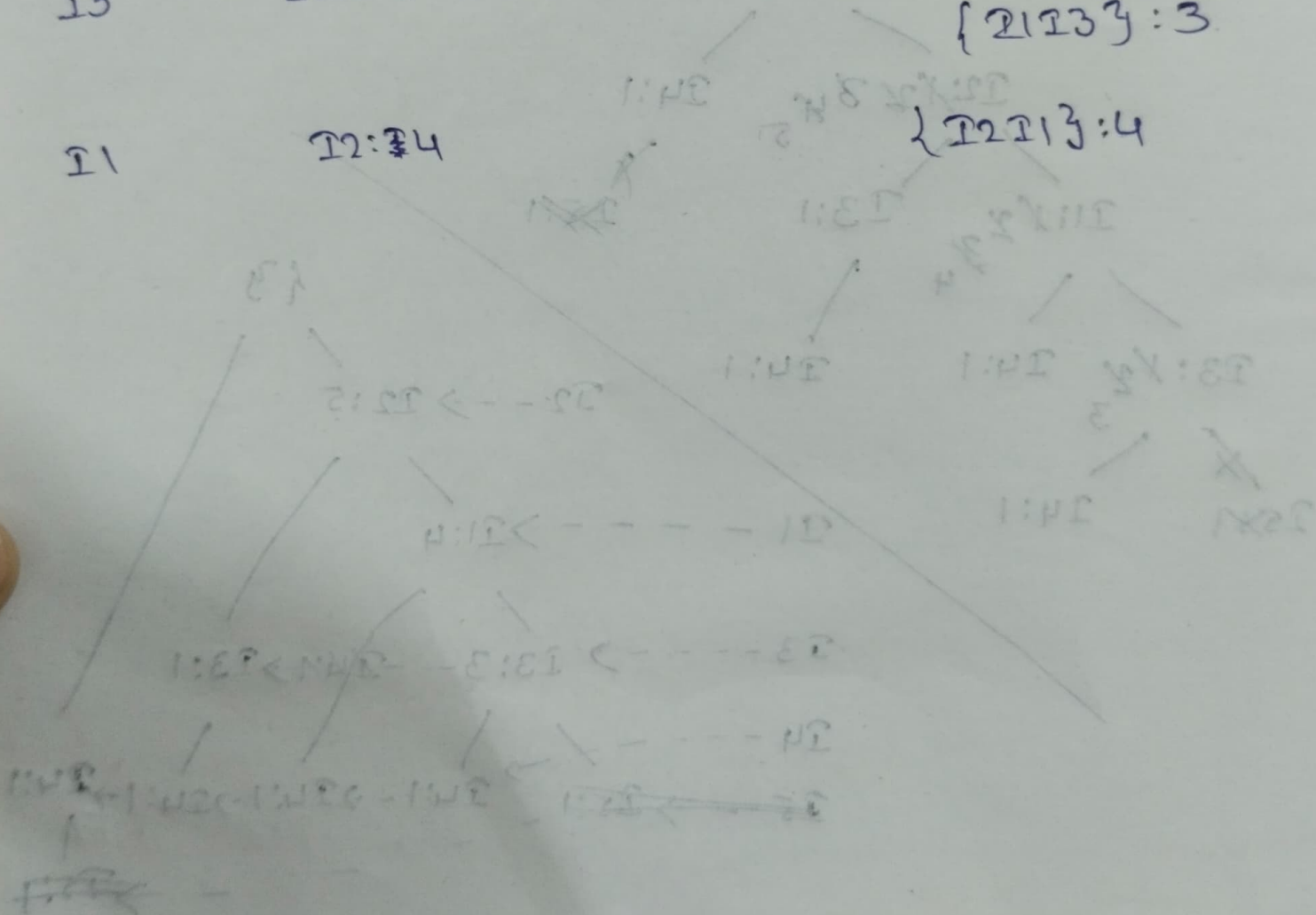
$I2I1:3$ $I1:1$

$\{I2I3\}:4$ $\{I2I1I3\}:3$
 $\{I1I3\}:3$

I1

$I2:4$

$\{I2I1\}:4$



2.

Given,

a)

 $m=2$ i.e., two classes

$$D=10$$

Yes = 5 tuples

No = 5 tuples

$$\text{Info}(D) = - \sum_{i=1}^m p_i \log p_i$$

$$= - \frac{5}{10} \log_2 \left(\frac{5}{10} \right) - \frac{5}{10} \log_2 \left(\frac{5}{10} \right)$$

$$= - \frac{5}{10} \times -1 - \frac{5}{10} \times -1$$

$$= \frac{5}{10} + \frac{5}{10}$$

$$= 1$$

computing expected information requirement for each attribute

Color

$$\text{Info}_{\text{Color}}(D) = \sum_{j=1}^J \frac{|D_j|}{|D|} \times \text{Info}(D)$$

Red Yellow Red

Yes 3 2

No 2 3

$$= \frac{5}{10} \times \left(- \frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$+ \frac{5}{10} \times \left(- \frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right)$$

$$= \frac{5}{10} (0.442 + 0.528) + \frac{5}{10} (0.442 + 0.528) = 0.97$$

$$\text{Gain}(\text{color}) = \text{Info}(D) - \text{Info}_{\text{color}}(D)$$

$$= 1 - 0.99$$

$$= 0.03$$

Type

$$\text{Gain}_{\text{type}}(D) = - \sum_{j=1}^V \frac{|D_j|}{D} \sum_{i=1}^V \text{Info}(D)$$

$$= \frac{6}{10} \times \left(-\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right) \begin{matrix} \text{Yes} & 4 & 1 \\ \text{No} & 2 & 3 \end{matrix}$$

$$+ \frac{4}{10} \times \left(-\frac{1}{4} \log_2 \left(\frac{1}{4} \right) - \frac{3}{4} \log_2 \left(\frac{3}{4} \right) \right)$$

$$= \frac{6}{10} \left(-\frac{4}{6} \times -0.584 - \frac{2}{6} \times -0.415 \right) + \frac{4}{10} \left(-\frac{1}{4} \times -2 - \frac{3}{4} \times -0.915 \right)$$

$$= \frac{6}{10} (0.389 + 0.138) + \frac{4}{10} (0.5 + 0.3118)$$

$$= 0.316 + 0.324$$

$$= 0.6408$$

$$\text{Gain}(\text{type}) = \text{Info}(D) - \text{Info}_{\text{type}}(D)$$

$$= 1 - 0.640$$

$$= 0.359$$

Origin :

Domestic

Imported

$$= \frac{5}{10} \left(-\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right) +$$

Yes

2

3

$$\frac{5}{10} \left(-\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.97$$

$$\text{Gain}(\text{origin}) = \text{Info}(D) - \text{Info}_{\text{origin}}(D)$$

$$= 1 - 0.97$$

$$= 0.03$$

$$\therefore \text{Gain}(\text{type}) = 0.359 \quad \checkmark \quad \text{Root node}$$

TYPE

sports

suV

color origin stolen

Red domestic yes

Red domestic no

Red domestic yes

yellow domestic no

yellow imported yes

Red imported yes

color origin stolen

yellow imported no

yellow imported yes

yellow domestic no

Red imported no

$$\begin{aligned} \text{Info}_{\text{sports}}(D) &= - \left[\frac{4}{6} \log\left(\frac{4}{6}\right) + \frac{2}{6} \log\left(\frac{2}{6}\right) \right] \\ &= - [-0.389 + -0.528] \\ &= 0.917 \end{aligned}$$

$$\text{Info}_{\text{color}}(D) = \frac{4}{6} I(3,2) + \frac{2}{6} I(1,1)$$

$$\begin{aligned} I(3,2) &= 0.871 \quad I(1,1) = 1 \quad I(3,1) = -\frac{3}{4} \log\left(\frac{3}{4}\right) \\ &\quad + \frac{1}{4} \log\left(\frac{1}{4}\right) \\ &= \frac{4}{6} (0.97) + \frac{2}{6} (1) = 0.884 \end{aligned}$$

$$\text{Info Gain}(\text{Color}) = 0.917 - 0.884 = 0.043. \quad \left. \begin{array}{l} = -0.311 \\ = 0.811 \end{array} \right\}$$

$$\text{Info}_{\text{origin}}(D) = \frac{4}{6} I(2,2) + \frac{2}{6} I(2,0)$$

$$I(2,2) = 1 \quad I(2,0) = \frac{2}{2} \log\left(\frac{2}{2}\right) = 0$$

$$= \frac{4}{6} (1) = 0.667$$

$$\begin{aligned} \text{Info Gain}(\text{Origin}) &= 0.917 - 0.667 = \\ &= 0.251 \end{aligned}$$

2 continue

$$\underline{\text{Info}(\text{sun})} = \begin{array}{l} 3 \text{ tuples no} \\ 1 \text{ tuple yes} \end{array}$$

$$\begin{aligned} -p_i \log p_i &= -\frac{1}{4} \log \frac{1}{4} - \frac{3}{4} \log \frac{3}{4} \\ &= -\frac{1}{4} \times -2 - \frac{3}{4} \times -0.415 \\ &= \frac{2}{4} + \frac{3}{4} \times 0.415 \end{aligned}$$

$$\text{Info}(\text{sun}) = 0.811$$

$$\text{Info}_{\text{color}}(D) = \sum_{j=1}^2 \frac{|D_j|}{|D|} \sum_{i=1}^2 \text{Info}(D_{ij})$$

	Yellow	Red
Yes	1	0
No	2	1

$$= \frac{1}{1} (-1 \log 1 + 0) + \frac{3}{4} \left(-\frac{2}{3} \log \frac{2}{3} - \frac{1}{3} \log \frac{1}{3} \right)$$

$$= 0 + \frac{3}{4} \left(-\frac{2}{3} \times -0.58 - \frac{1}{3} \times -1.58 \right)$$

$$= 0.685$$

$$\begin{aligned} \text{Info gain}(\text{color}) &= 0.811 - 0.685 \\ &= 0.126 \end{aligned}$$

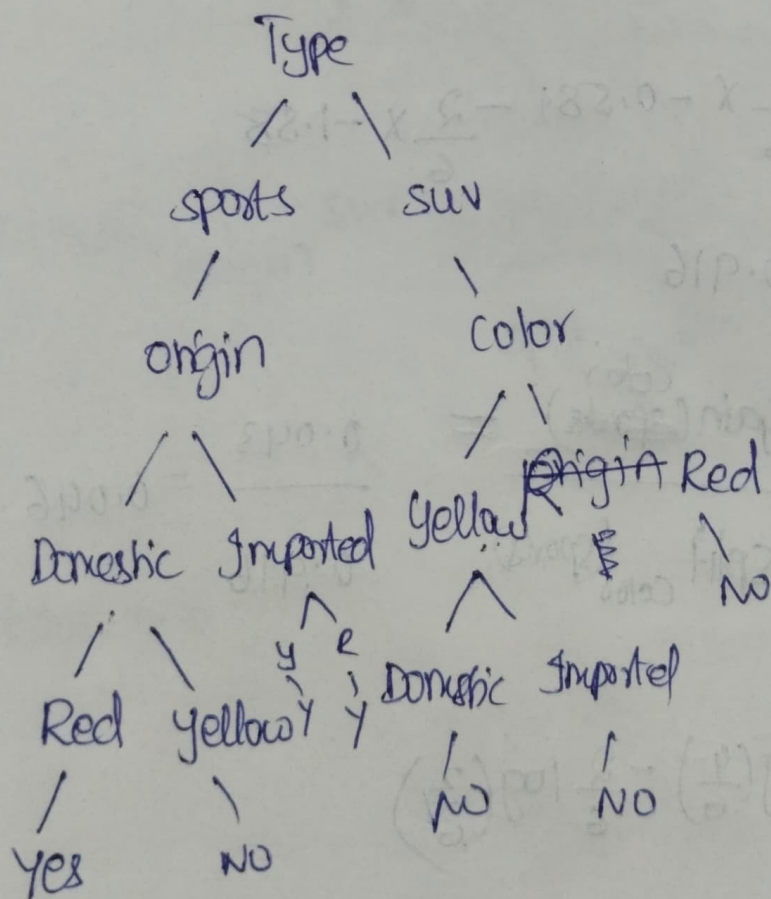
origin

$$\text{Info origin (D)} = 0.685$$

$$\text{Gain(origin)} = 0.126$$

	Imported	Domestic
yes	1	0
NO	2	1

Decision tree



$$b) \text{ gain ratio (A)} = \frac{\text{gain(A)}}{\text{splitInfo(A)}}$$

$$\text{splitInfo}_{(\text{type})}(\text{D}) = - \sum_{j=1}^{\checkmark} \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$= - \frac{6}{10} \times \log_2 \left(\frac{6}{10} \right) - \frac{4}{10} \times \log_2 \left(\frac{4}{10} \right)$$

$$= - \frac{6}{10} \times -0.736 - \frac{4}{10} \times -1.321$$

$$= 0.441 + 0.528 = 0.9696$$

$$\text{gain ratio (type)} = \frac{\text{gain (Type)}}{\text{split info Type}^{(D)}} = \frac{0.359}{0.9696} = 0.370$$

$$b) \text{split info}_{(\text{color})}(D) = - \sum_{j=1}^{\sqrt{}} \frac{|D_j|}{|D|} \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$= - \frac{5}{10} \log \frac{5}{10} - \frac{5}{10} \log \frac{5}{10}$$

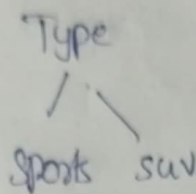
$$= 1$$

$$\text{gain ratio}(\text{color}) = \frac{\text{gain}(\text{color})}{\text{split info}(\text{color})} = \frac{0.03}{1} = 0.03$$

$$c) \text{split info}_{(\text{origin})}(D) = - \frac{5}{10} \log \frac{5}{10} - \frac{5}{10} \log \frac{5}{10} = 1$$

$$\text{gain ratio} = 0.03$$

∴ gain ratio for type is none



gain ratio for sports

$$\begin{aligned}
 \text{Gain(sports)} &= -\frac{4}{6} \log \frac{4}{6} - \frac{2}{6} \log \frac{2}{6} \\
 &= -\frac{4}{6} \times -0.584 - \frac{2}{6} \times -1.58 \\
 &= 0.916
 \end{aligned}$$

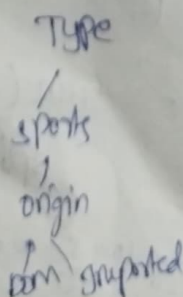
color

$$\text{gain ratio}_{\text{color}}^{\text{sports}} = \frac{\text{Gain}_{\text{color}}^{\text{sports}}}{\text{Split}_{\text{color}}^{\text{sports}}} = \frac{0.043}{0.916} = 0.046$$

$$\begin{aligned}
 \text{split}_{\text{color}}^{\text{sports}} &= -\frac{4}{6} \log \left(\frac{4}{6} \right) - \frac{2}{6} \log \left(\frac{2}{6} \right) \\
 &= 0.916
 \end{aligned}$$

$$\begin{aligned}
 \text{gain ratio}_{\text{origin}}^{\text{sports}} &= \frac{\text{Gain}_{\text{origin}}^{\text{sports}}}{\text{split}_{\text{origin}}^{\text{sports}}} = \frac{0.251}{0.916} = 0.274
 \end{aligned}$$

So origin > color



gain ratio for suv

color

$$\text{gain ratio}_{\text{color}}^{\text{suv}} = \frac{\text{gain}(\text{color})}{\text{split}_{\text{color}}^{\text{suv}}} = \frac{0.126}{0.811} = 0.155$$

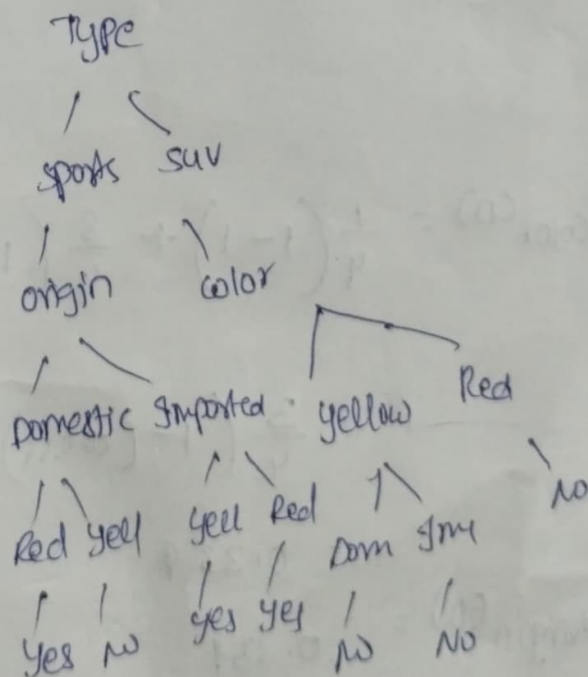
$$= -\frac{3}{4} \log\left(\frac{3}{4}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right)$$

$$= -\frac{3}{4} \times -0.415 - \frac{1}{4} \times -2$$

$$= 0.811$$

$$\text{gain ratio}_{\text{origin}}^{\text{suv}} = \frac{\text{gain}(\text{origin})}{\text{split}_{\text{orig}}^{\text{suv}}} = \frac{0.126}{0.811} = 0.155$$

Decision tree



c) Gini Impurity Index:

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2 \quad p_i = \frac{|C_i(D)|}{|D|}$$

$$Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2)$$

$$\begin{aligned} Gini_{type}(D) &= \frac{6}{10} \times \left(1 - \left(\left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right) \right) + \frac{4}{10} \times \left(1 - \left(\left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right) \right) \\ &= \frac{6}{10} \times \left(1 - \left(\frac{16}{36} + \frac{4}{36} \right) \right) + \frac{4}{10} \times \left(1 - \left(\frac{1}{16} + \frac{9}{16} \right) \right) \\ &= \frac{6}{10} \times \left(1 - \frac{20}{36} \right) + \frac{4}{10} \times \left(1 - \frac{10}{16} \right) \\ &= 0.6 \times 0.444 + 0.4 \times 0.375 \\ &= 0.446 \end{aligned}$$

$$\begin{aligned} Gini(D) &= 1 - \left(\left(\frac{5}{10} \right)^2 + \left(\frac{5}{10} \right)^2 \right) \\ &= 0.5 \end{aligned}$$

$\Delta =$

same for $Gini_{origin}(D) = 0.48$

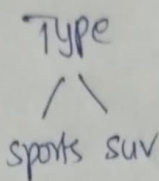
Take small value in Gini, so Type is best

$$\begin{aligned} Gini_{color}(D) &= \frac{5}{10} \left(1 - \left(\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right) \right) \\ &+ \frac{5}{10} \left(1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) \right) \\ &= \end{aligned}$$

$$\begin{aligned} &= \frac{5}{10} \left(1 - (0.36 + 0.16) \right) + \\ &\frac{5}{10} \left(1 - (0.36 + 0.16) \right) \end{aligned}$$

$$= \frac{5}{10} \times 0.48 + \frac{5}{10} \times 0.48 = 0.48$$

Q) Gini Impurity



sports

$$Gini_{color}(D) = \frac{4}{6} \left(1 - \left(\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right) \right) + \frac{2}{6} \left(1 - \left(\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right) \right)$$

$$= \frac{4}{6} (1 - (0.625)) + \frac{2}{6} (1 - 0.5)$$

$$= 0.667 (0.375) + 0.167$$

$$= 0.417$$

$$Gini_{origin}(D) = 0.417$$

suv

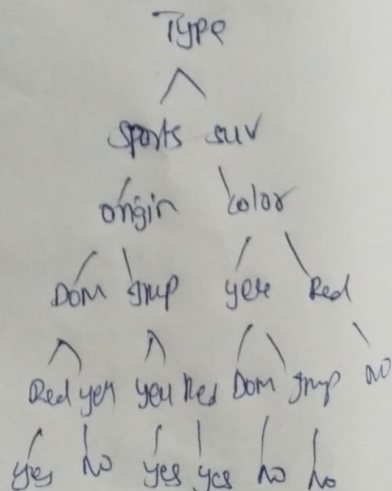
$$Gini_{color}(D) = \frac{1}{4} (1 - 1) + \frac{3}{4} \left(1 - \left(\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right) \right)$$

$$= \frac{3}{4} (1 - (0.555))$$

$$= 0.334$$

$$Gini_{origin}(D) = 0.334$$

Now the decision tree is



3. Naive Bayes's method

give birth	can fly	live in water	Have legs	class
yes	no	yes	no	?

Solution

To find which class maximizes $P(x|c_i)P(c_i)$

$$D = 20 \begin{array}{l} \nearrow \text{mammals} = 7 \\ \searrow \text{non-mammals} = 13 \end{array}$$

$$P(\text{mammals}) = \frac{7}{20} = 0.35$$

$$P(\text{non-mammals}) = \frac{13}{20} = 0.65$$

$$P(x|c_i)P(c_i) = \cancel{P(c_i)} \times \cancel{P(x)}$$

$$P(x|c_i) = \prod_{k=1}^n P(x_k|c_i)$$

now

$$P(\text{give birth} = \text{yes} \mid \text{class} = \text{mammals}) = \frac{6}{7} = 0.857$$

$$P(\text{give birth} = \text{yes} \mid \text{class} = \text{non-mammals}) = \frac{1}{13} = 0.076$$

$$P(\text{can fly} = \text{no} \mid \text{class} = \text{mammals}) = \frac{6}{7} = 0.857$$

$$P(\text{can fly} = \text{no} \mid \text{class} = \text{non-mammals}) = \frac{10}{13} = 0.769$$

$$P(\text{live in water} = \text{yes} \mid \text{class} = \text{mammals}) = \frac{2}{7} = 0.285$$

$$P(\text{live in water} = \text{yes} \mid \text{class} = \text{non-mammals}) = \frac{3}{13} = 0.230$$

$$P(\text{have legs} = \text{no} \mid \text{class} = \text{mammals}) = \frac{2}{7} = 0.285$$

$$P(\text{have legs} = \text{no} \mid \text{class} = \text{non-mammals}) = \frac{4}{13} = 0.30$$

$$P(X \mid \text{class} = \text{mammals}) = 0.857 \times 0.857 \times 0.285 \times 0.285 = 0.059$$

$$P(X \mid \text{class} = \text{non-mammals}) = 0.077 \times 0.76 \times 0.230 \times 0.3 = 0.0041$$

$$P(X \mid \text{class} = \text{mammals}) \times P(\text{mammals}) = 0.059 \times 0.35 = 0.020$$

$$\begin{aligned} P(X \mid \text{class} = \text{non-mammals}) \times P(\text{non-mammals}) &= 0.65 \times 7.5 \times 10^{-4} \times 0.041 \\ &= 4.29 \times 10^{-4} \\ &= 0.00026 \end{aligned}$$

$\therefore \text{class} = \text{mammals}$

4

Select randomly 3 objects = T_1, T_6, T_{11}

	C_1 T_1	C_2 T_6	C_3 T_{11}	cluster
T_1	0	3	1	C_1
T_2	2	2	1	C_3
T_3	3	0	3	C_2
T_4	3	2	2	C_2
T_5	1	2	1	C_1
T_6	3	0	3	C_2
T_7	1	3	2	C_1
T_8	3	0	3	C_2
T_9	2	2	1	C_3
T_{10}	2	2	1	C_3
T_{11}	1	3	0	C_3
T_{12}	2	1	2	C_2
T_{13}	2	3	1	C_3
T_{14}	2	1	3	C_2

 $C_1 \Rightarrow T_5, T_7, T_{11}$

$C_2 \Rightarrow T_3, T_4, T_6, T_8, T_{12}, T_{14}$

$C_2 \Rightarrow$ ~~T_1~~ , $T_2, T_9, T_{10}, T_{11}, T_{13}$

C_1 (Telugu, M, ~~Mech~~)

$C_2 \Rightarrow T_3 = \text{Tamil, F, ECE}$

$T_4 = \text{Hin, F, CSE}$

$T_6 = \text{Tamil, F, ECE}$

$T_8 = \text{Tamil, F, ECE}$

$T_{12} = \text{Tamil, M, CSE}$

$T_{14} = \text{Tamil, F, Mech}$

$C_2 \Rightarrow \text{Tamil, F, ECE}$

C_1

$C_3 \Rightarrow$

~~$T_1 = \text{Tel, M, Mech}$~~

$T_2 = \text{Tamil, M, CSE}$

$T_9 = \text{Tel, F, CSE}$

$T_{10} = \text{Tam, M, CSE}$

$T_{11} = \text{Tel, M, CSE}$

$T_{13} = \text{Hin, M, CSE}$

$C_3 \Rightarrow \text{Tel, M, CSE}$

	(Telugu, C1, Mech) C1	C2 (Tum, F, ECE)	C3	Clu
T1	0	-	-	C1
T2	2	2	0	C3
T3	3	0	-	C2
T4	3	2	2	C2
T5	1	2	1	C1
T6	3	0	-	C2
T7	1	3	2	C1
T8	3	0	-	C2
T9	2	2	2	C3
T10	1	2	0	C3
T11	2	3	1	C3
T12	1	1	1	C2
T13	2	3	1	C3
T14	1	1	2	C2

C1: T1, T5, T7

C2: T3, T4, T6, T8, T12, T14

C3: T2, T9, T10, T11, T13

5. K-means clustering

Converged c_3 (Tamil, M)

	d_1 (2,1,0)	d_2 (5,8)	d_3 (1,2)	cluster
$A_1(2,1,0)$	0	8.60	8.06	C_1
$A_2(2,1,5)$	5	4.24	3.16	C_3
$A_3(8,1,4)$	8.48	5	7.28	C_2
$A_4(5,1,8)$	-	0	-	C_2
$A_5(7,1,5)$	7.07	8.16	-	C_2
$A_6(6,1,4)$	7.21	4.12	5.38	C_2
$A_7(6,1,2)$	-	-	0	C_3
$A_8(4,1,9)$	2.2	1.41	-	C_2

$$\sqrt{2^2+1}$$

$$\sqrt{3^2+2^2} = 3.60, \sqrt{1^2+8^2} = 8.06$$

$$\sqrt{3^2+3^2} = 4.24, \sqrt{10} = 3.16$$

$$\sqrt{6^2+6^2} = 8.48, \sqrt{3^2+4^2} = 5, \sqrt{7^2+2^2} = 7.28$$

$$\sqrt{4^2+6^2} =$$

$$\sqrt{5^2+8^2} = 7.07$$

$$\sqrt{2^2+3^2}$$

$$C_1 = A_1$$

$$C_2 = A_3, A_4, A_5, A_6, A_8$$

$$C_3 = A_2, A_7$$

$$C_1 \text{ new centroid} = A_1 = (2, 1, 1, 0)$$

$$C_2 = (6, 6)$$

$$X = \frac{(8 + 5 + 7 + 6 + 4)}{5} = 6$$

$$Y = \frac{(4 + 8 + 5 + 4 + 9)}{5} = 6$$

$$C_3 X = \frac{(2 + 1)}{2} = 1.5$$

$$C_3 Y = \frac{(5 + 2)}{2} = 3.5$$

$$C_3 (1.5, 3.5)$$

	C_1	C_2	C_3	cluster/new clusters	
Datapoints	$(2, 1, 1, 0)$	$(6, 6)$	$(1.5, 3.5)$		
$A_1 (2, 1, 1, 0)$	0	-	-	C_1	C_1
$A_2 (2, 1, 5)$	5	4.12	1.58	C_3	C_3
$A_3 (8, 4)$	8.48	2.82	-	C_2	C_2
$A_4 (5, 8)$	3.60	3.16	6.36	C_2	C_2
$A_5 (7, 5)$	7.07	1.414	-	C_2	C_2
$A_6 (6, 4)$	7.21	2	4.5	C_2	C_2
$A_7 (1, 2)$	8.06	6.4	1.58	C_3	C_3
$A_8 (4, 9)$	2.2	3.60		C_2	C_1



Not get converged find new centroids

$$A_1 \left(\frac{2+4}{2}, \frac{10+9}{2} \right) = (3, 9.5)$$

$$B_1 \left(\frac{8+5+7+6}{4}, \frac{4+8+5+4}{4} \right) = 6.5, 5.25$$

$$C_1 (1.5, 3.5)$$

⑥ Given,

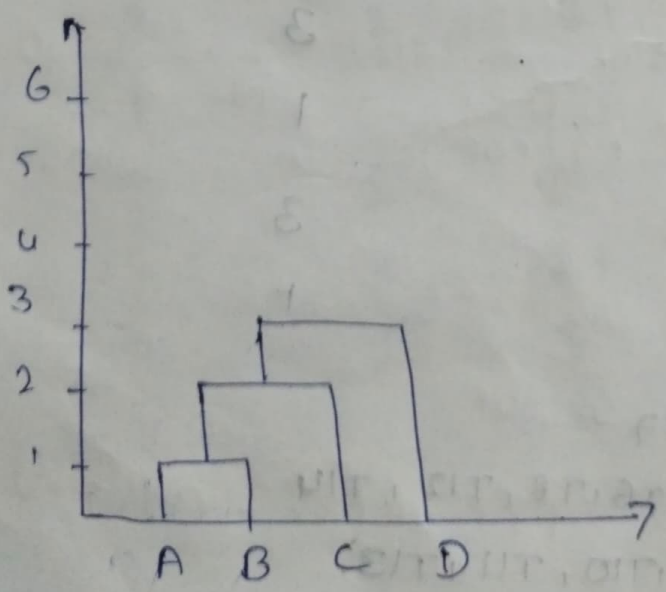
	A	B	C	D
A	0			
B	①	0		
C	4	2	0	
D	5	6	3	0

Single Link

	A, B	C	D
A, B	0		
C	②	0	
D	5	3	0

	A, B, C	D
A, B, C	0	
D	③	0

Dendrogram



Complete link

	A	B	C	D
A	0			
B	①	0		
C	4	2	0	
D	5	6	3	0

	A, B	C	D
A, B	0		
C	4	0	
D	6	③	0

A, B, D C, D, F

	A, B, D	C, D, F
A, B, D	0	
C	⑥	0

Dendrogram

